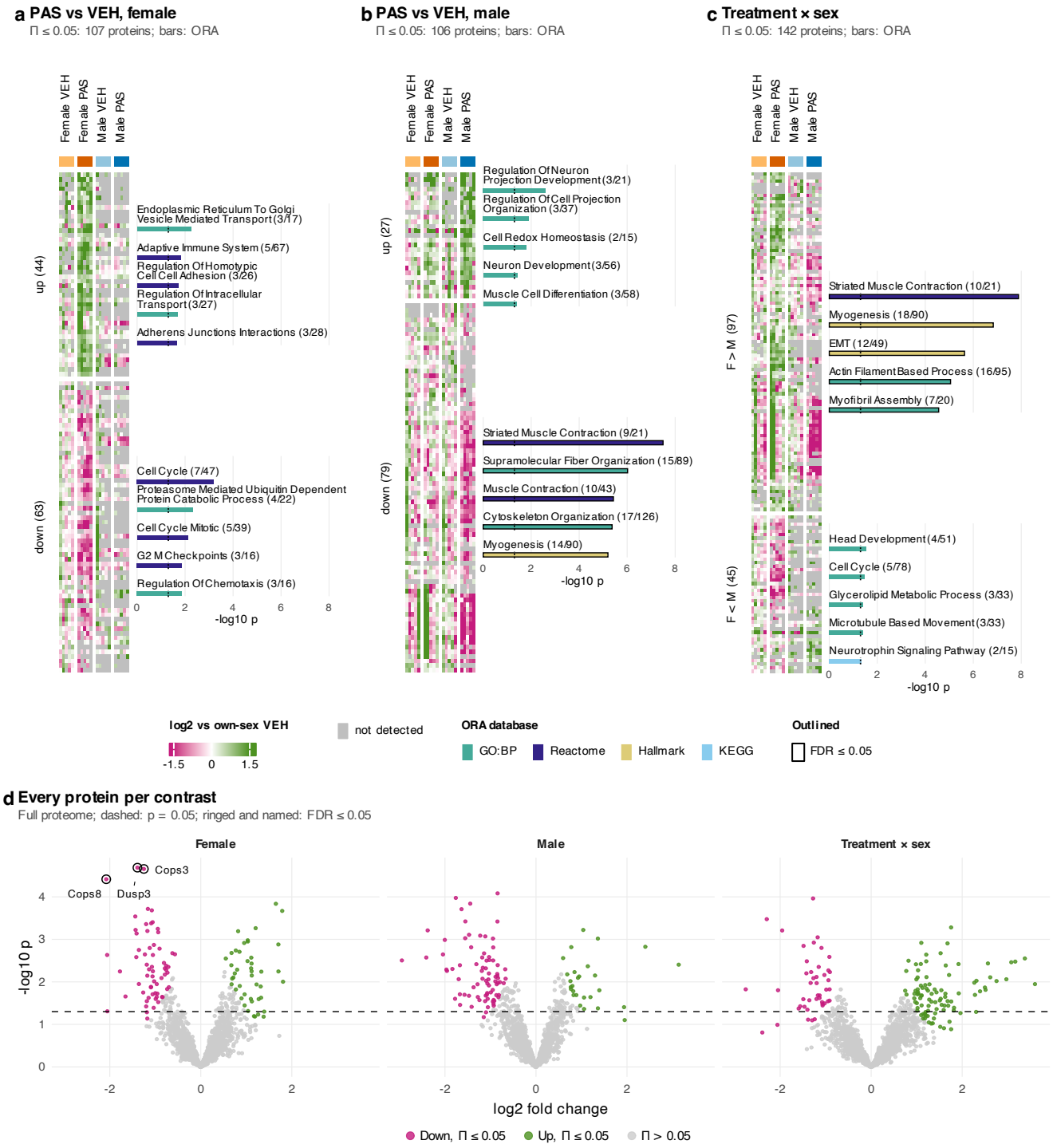

a PAS vs VEH, female
Π ≤ 0.05: 107 proteins; bars: ORA
Female VEH
Female PAS
Male VEH
Male PAS
up (44)
down (63)
Endoplasmic Reticulum To Golgi Vesicle Mediated Transport (3/17)
Adaptive Immune System (5/67)
Regulation Of Homotypic Cell Cell Adhesion (3/26)
Regulation Of Intracellular Transport (3/27)
Adherens Junctions Interactions (3/28)
Cell Cycle (7/47)
Proteasome Mediated Ubiquitin Dependent Protein Catabolic Process (4/22)
Cell Cycle Mitotic (5/39)
G2 M Checkpoints (3/16)
Regulation Of Chemotaxis (3/16)
0 2 4 6 8
-log10 p
b PAS vs VEH, male
Π ≤ 0.05: 106 proteins; bars: ORA
up (27)
down (79)
Regulation Of Neuron Projection Development (3/21)
Regulation Of Cell Projection Organization (3/37)
Cell Redox Homeostasis (2/15)
Neuron Development (3/56)
Muscle Cell Differentiation (3/58)
Striated Muscle Contraction (9/21)
Supramolecular Fiber Organization (15/89)
Muscle Contraction (10/43)
Cytoskeleton Organization (17/126)
Myogenesis (14/90)
0 2 4 6 8
-log10 p
c Treatment × sex
Π ≤ 0.05: 142 proteins; bars: ORA
F > M (97)
F < M (45)
Striated Muscle Contraction (10/21)
Myogenesis (18/90)
EMT (12/49)
Actin Filament Based Process (16/95)
Myofibril Assembly (7/20)
Head Development (4/51)
Cell Cycle (5/78)
Glycerolipid Metabolic Process (3/33)
Microtubule Based Movement (3/33)
Neurotrophin Signaling Pathway (2/15)
0 2 4 6 8
-log10 p
log2 vs own-sex VEH
-1.5 0 1.5
not detected
ORA database
GO:BP
Reactome
Hallmark
KEGG
Outlined
FDR ≤ 0.05
d Every protein per contrast
Full proteome; dashed: p = 0.05; ringed and named: FDR ≤ 0.05
Female
Male
Treatment × sex
Cops3
Cops8
Dusp3
-log10 p
0 1 2 3 4
-2 0 2
log2 fold change
Down, Π ≤ 0.05
Up, Π ≤ 0.05
Π > 0.05